

Predicting Insurance Costs

NSDS Mini - Datathon 2025



Problem framing & objective

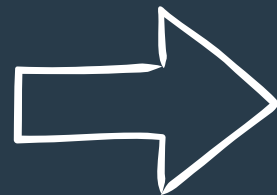
Problem:

The rising costs and complexities in the healthcare sector create an urgent need for efficient predictive models to anticipate medical insurance prices accurately.

Objectives:

We aim to :

- Interpret and analyze data to uncover valuable information and actionable insights.
- Make use of feature engineering to support data-driven decision-making.
- Apply and compare machine learning techniques.



To forecast medical insurance charges, providing insights for pricing strategies and risk management.



Exploratory Data Analysis (EDA)

- No missing values → dataset is generally clean; minimal preprocessing needed.
- “Charges” is strongly right-skewed → a few very high bills pull the mean up.
- Log-transforming charges makes the distribution more normal
→ better for linear models/correlation analysis.

Smoker & BMI validity vs US benchmark

- **Scope & sample:** 1,338 records. We sanity-check smoker share and BMI distribution against external U.S. benchmarks (high level).
- **Smoking rate:** Dataset $\approx$ 20% smokers → plausible and within typical national ranges, depending on the year/definition, so acceptable for modeling.
- **Obesity:** $\sim$ 50% BMI $\geq$ 30 vs $\sim$ 40% benchmark → +10 pp higher. This likely reflects the sample mix (region, age, etc.).
- **Representativeness & size:** With $n = 1338$, some deviation from population rates is expected; results are reasonable for modeling but not for precise population inference.

Group	Count	Proportion
Non-smokers	1064	79.5%
Smokers	274	20.5%
Total	1338	100%

Fig 1: Proportion of “smoker”

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

Fig 2: Main features description

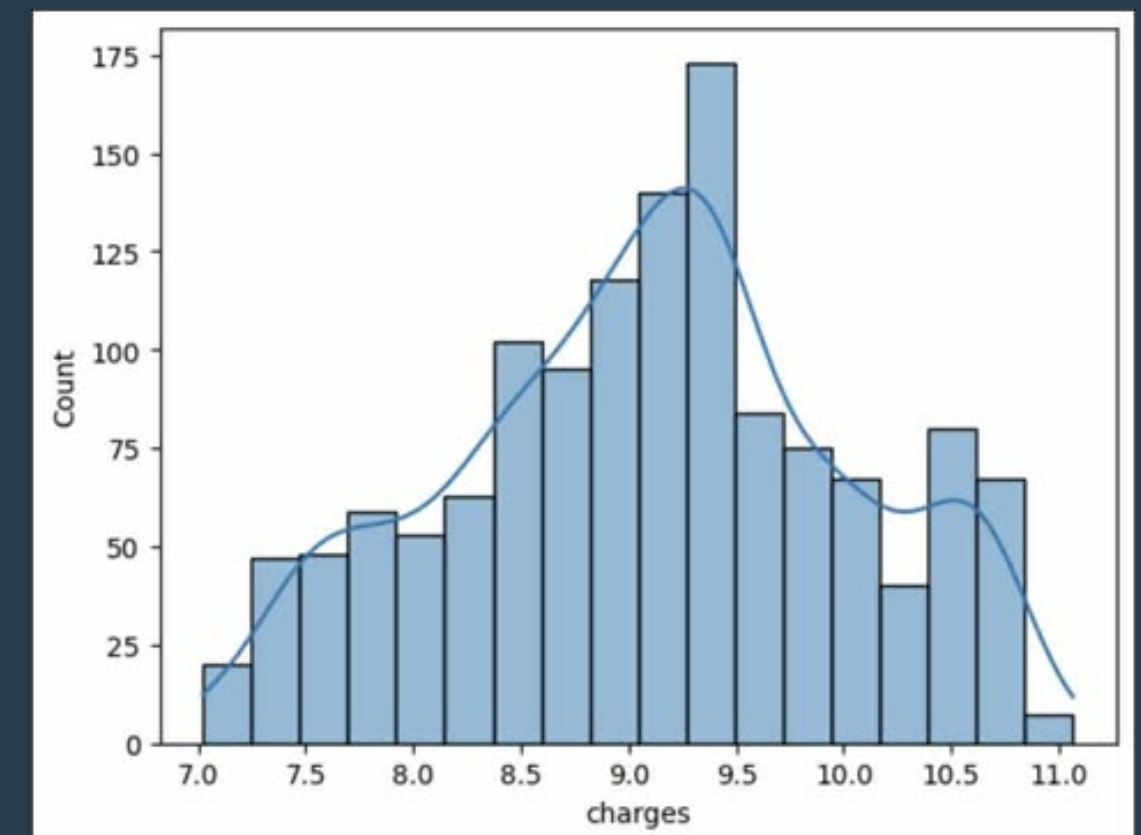


Fig 3: Histogram of charges under natural log (more normal)

Feature-level Insights

Key observations:

- **Age:** charges increase with age, leading older customers to pay more.
- **BMI:** higher BMI → somewhat higher charges; very high BMI is more expensive.
- **Children:** very weak effect.
- **Region/sex:** little or no effect → likely low importance.

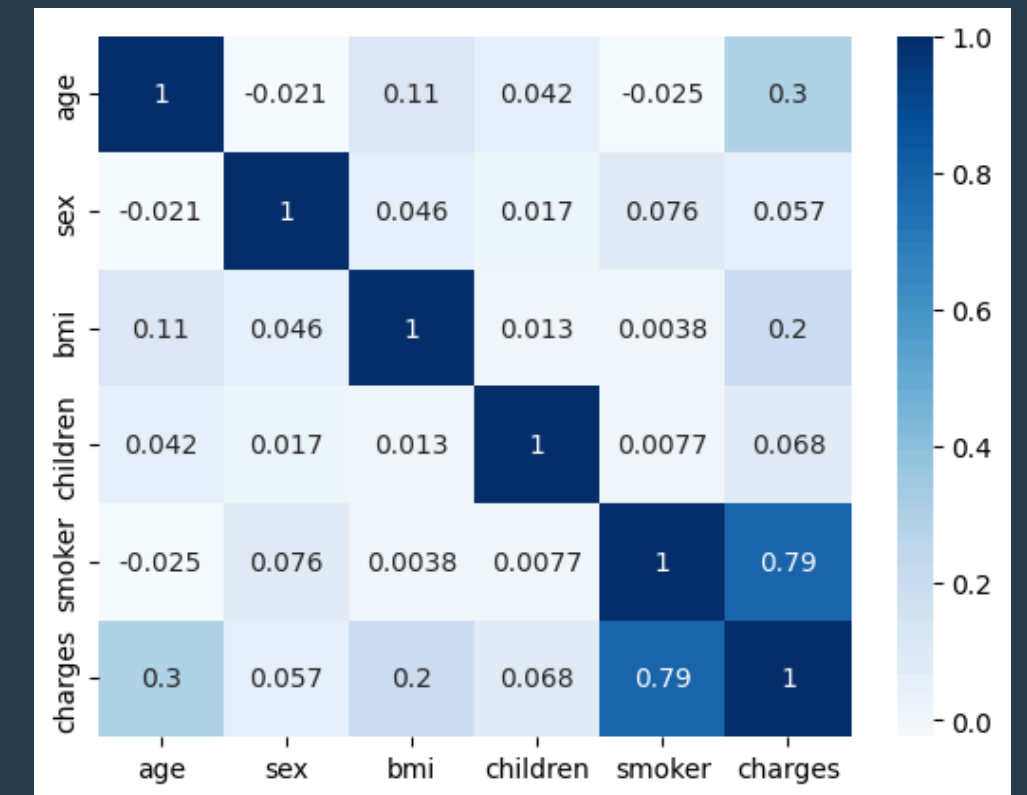


Fig 5: Correlation Heatmap

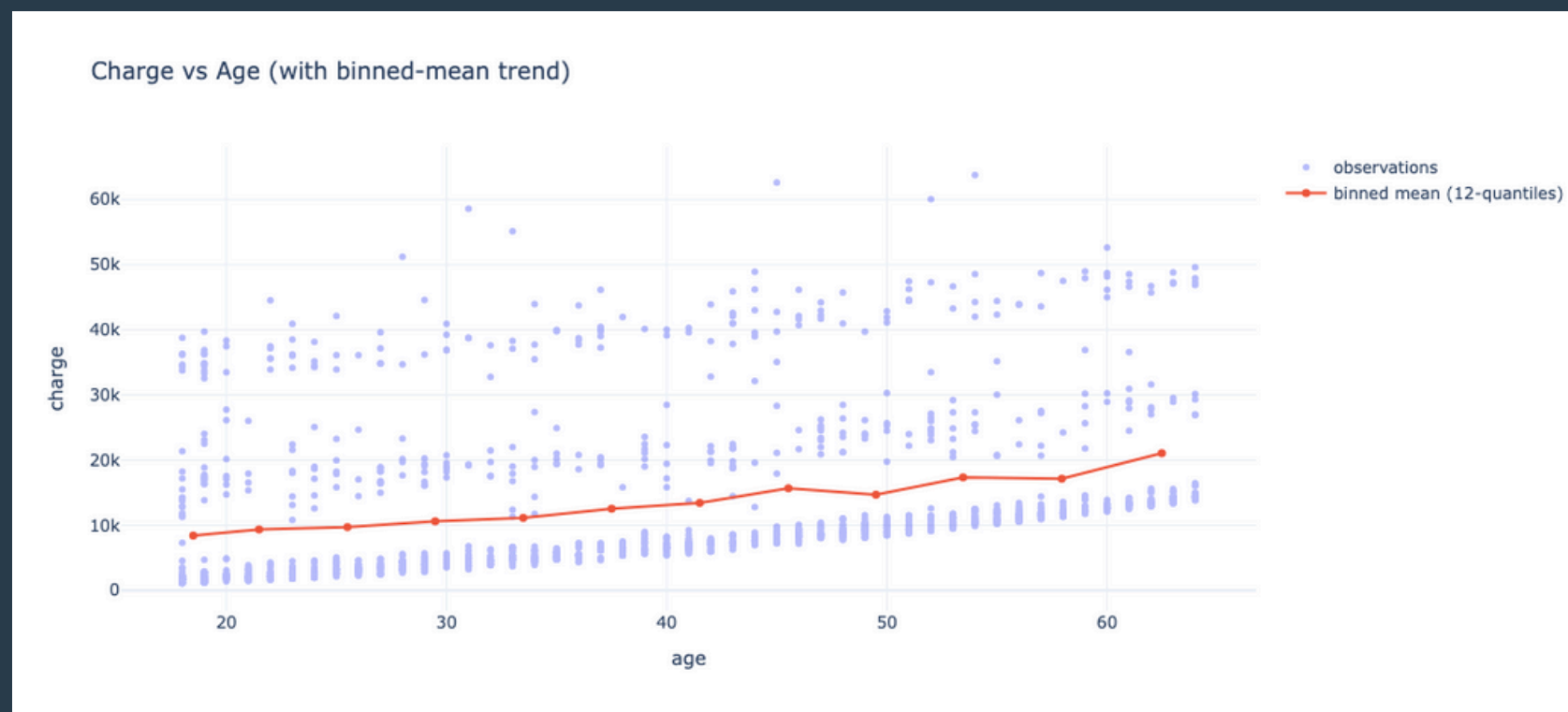


Fig 4: Charges vs Age

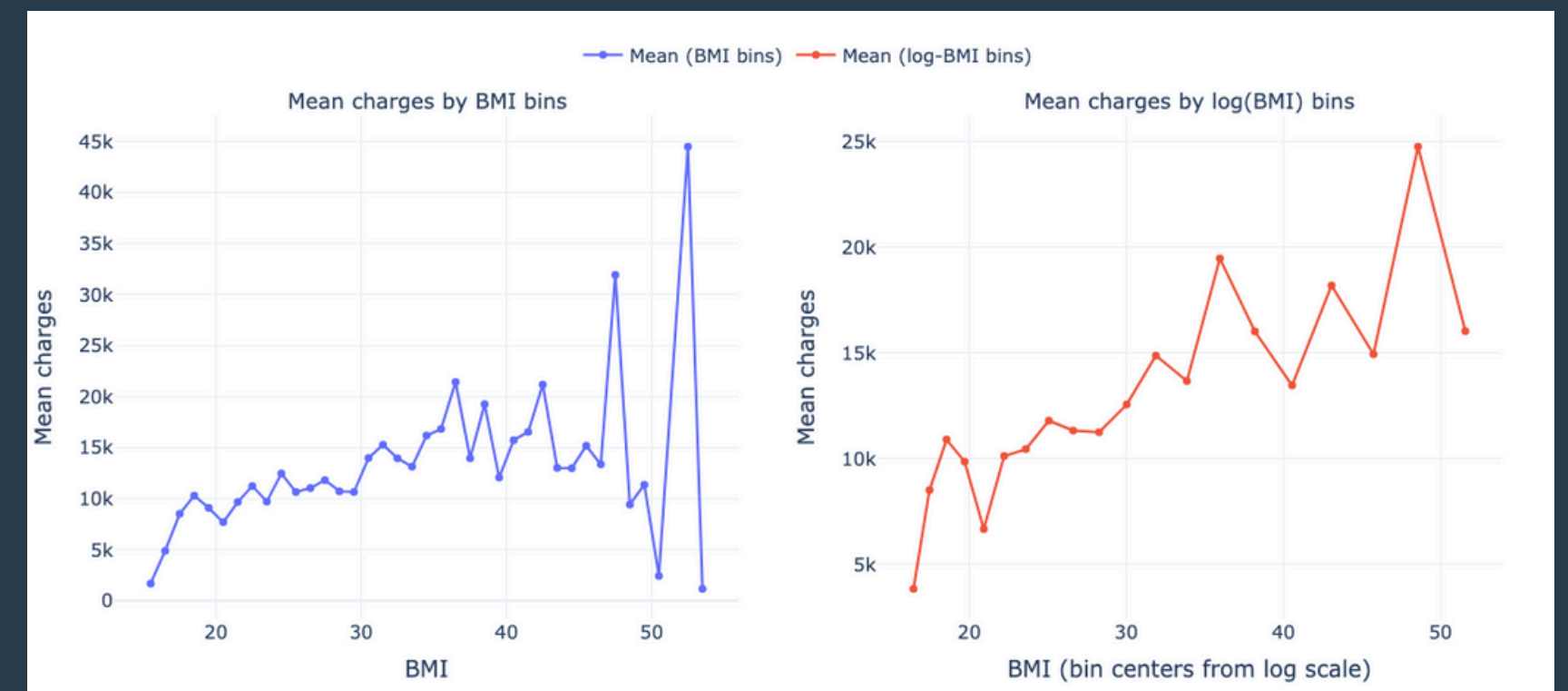


Fig 6: Charges vs BMI

Impact of Smoking on Charges

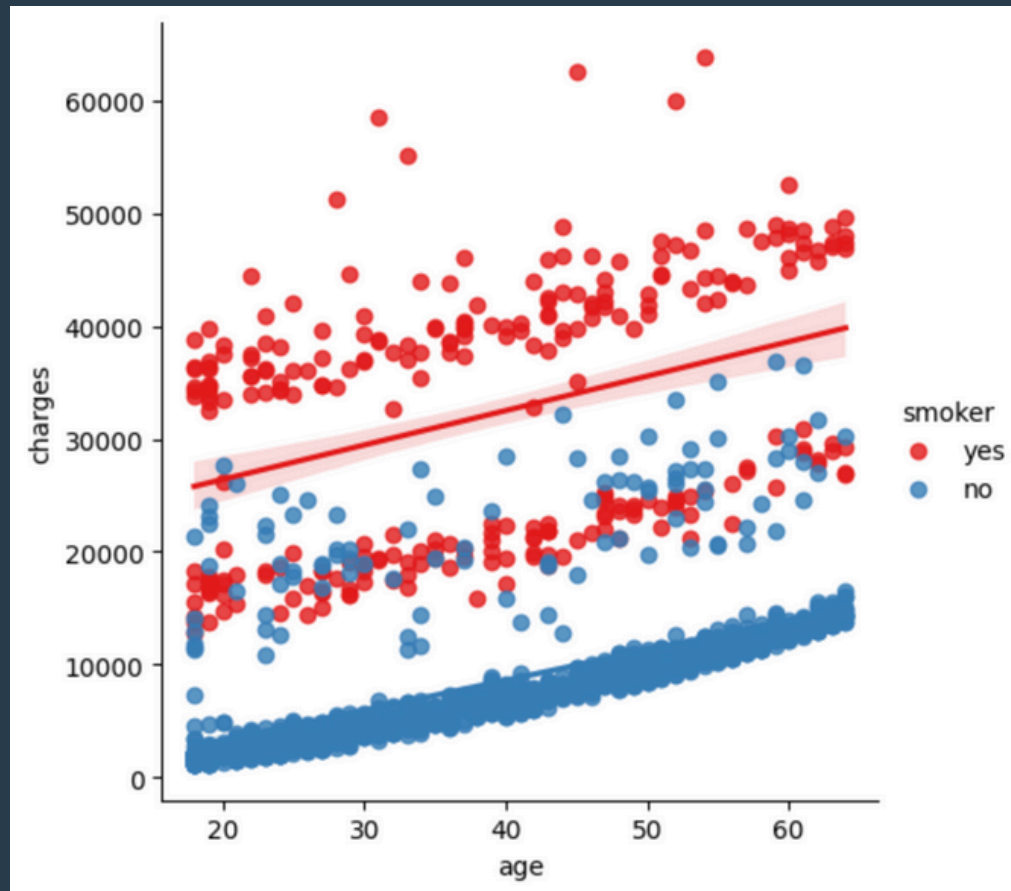


Fig 7: Charges by age

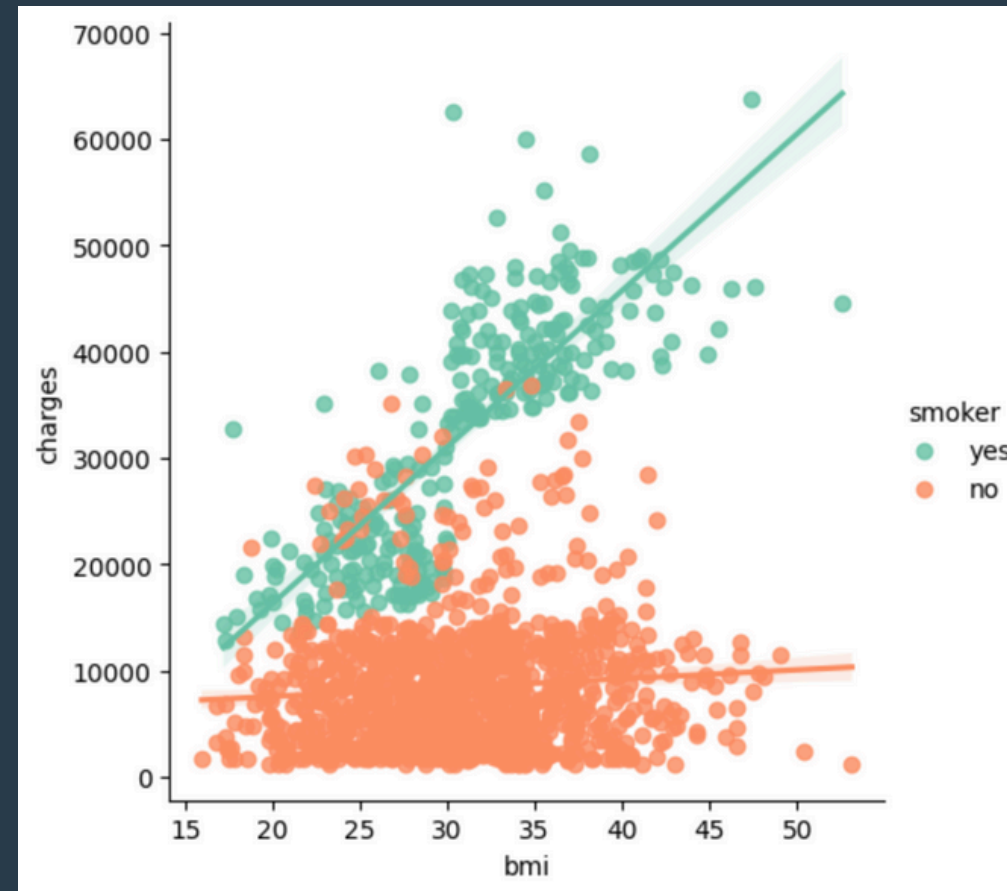


Fig 8: Charges by BMI

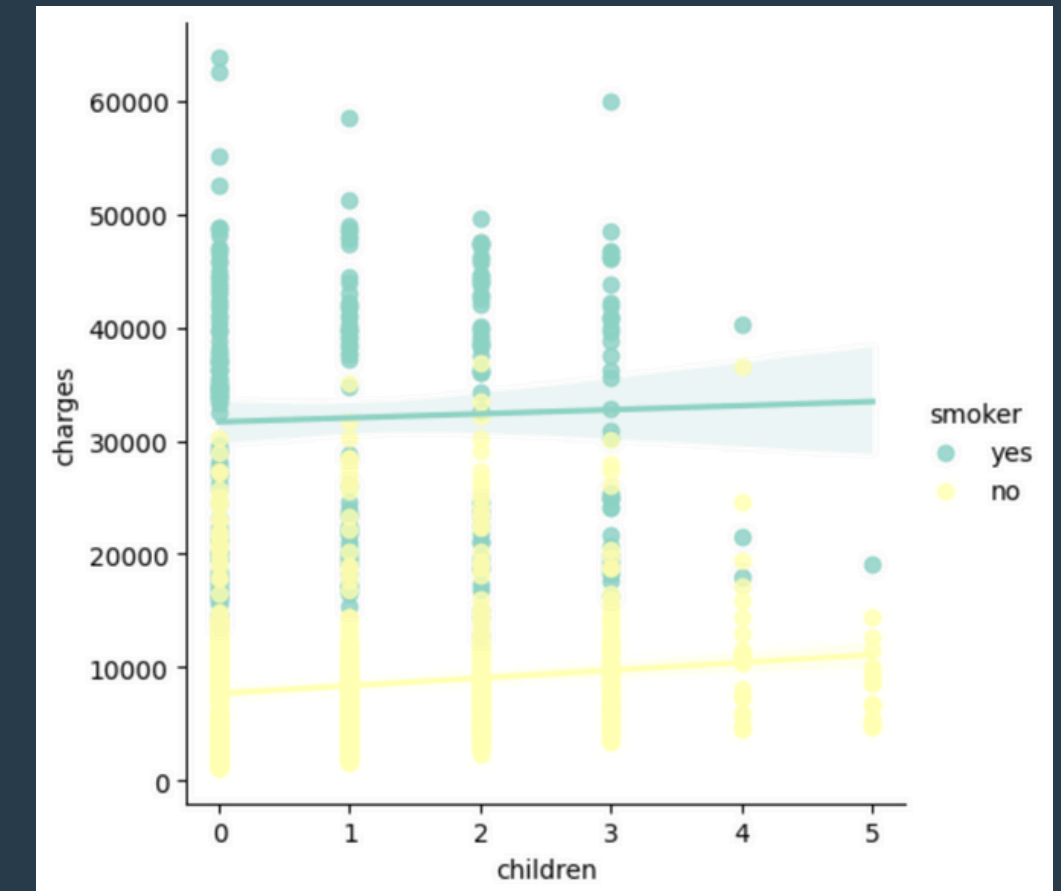


Fig 9: Charges by children

At every age, **smokers' charges** are higher.

High-BMI smokers → higher charges.

Children effect is flat vs. smoking effect.

- Smoking dwarfs other effects; smokers pay dramatically more at any age/BMI.
- BMI × smoking interaction for high-BMI smokers is the most expensive group.
- “Children” adds little compared to smoking.

Avg Charges:

- Smoker: **\$32050.231832**
- Non-smoker: **\$8434.268298**

Feature Engineering

Based on the given variables, we decided to add new features to support machine learning approaches, in which:

- **Age_Group** & **bmi_class** (interpretable risk segments)
- Centered continuous: **age_c**, **bmi_c** = $x - \text{mean}(x)$
→ stabilizes learning.
- Curvature terms: **age_sq**, **bmi_sq**
→ capture non-linear effects.
- Threshold/hinge: **age_h59**, **bmi_h25/30/35** = $\max(x - \text{cutoff}, 0)$
→ measures “excess risk” above cutoffs.
- Interactions: **smoker×age**, **smoker×bmi** (and with hinges), **age×bmi**
→ models amplified combined risks.
- Risk cell + credibility: **risk_cell** = Age_Group | bmi_class | smoker;
- **cell_mean_cred** = $w * \text{cell_mean} + (1 - w) * \text{overall}$,
with $w = n / (n + 30)$ → robust group signal.

7	Age_Group	1338	non-null	int64
8	bmi_class	1338	non-null	int64
9	age_c	1338	non-null	float64
10	bmi_c	1338	non-null	float64
11	age_sq	1338	non-null	float64
12	bmi_sq	1338	non-null	float64
13	age_h59	1338	non-null	int64
14	bmi_h25	1338	non-null	float64
15	bmi_h30	1338	non-null	float64
16	bmi_h35	1338	non-null	float64
17	smoker_raw	1338	non-null	int64
18	sex_raw	1338	non-null	int64
19	smoker_x_age	1338	non-null	Int64
20	smoker_x_bmi	1338	non-null	Float64
21	age_x_bmi	1338	non-null	float64
22	smoker_x_age_h59	1338	non-null	Int64
23	smoker_x_bmi_h30	1338	non-null	Float64
24	risk_cell	1338	non-null	int64
25	cell_mean_cred	1338	non-null	float64

Fig 10: Series of new features in the dataset

Regression & Modeling Approaches

Models tested (tuned):

- **Linear family** (Linear, Ridge/Lasso/ElasticNet, BayesianRidge, Huber, Tweedie)
- **Kernel/Neighbors** (SVR-RBF, KNN), **Neural** (MLP)
- **Trees/Ensembles** (Decision Tree, Random Forest, ExtraTrees, GradientBoosting, HistGB, AdaBoost) + **Baseline** (mean).

Train/Test split: 80% training, 20% holdout test (same split used for all models), no cross-validation.

Metrics reported (on test set): MAE, MSE, R^2 (**lower** MAE/MSE and **higher** R^2 are better).

METRICS	MAE	MSE	R^2
 GradientBoosting	1543.6810	4482.6544	0.8621
HuberRegressor	2008.6410	5067.4036	0.8238
RandomForest	2430.2287	4380.5834	0.8683

Fig 11: Model Performance



Remarks on Gradient Boosting:

- Best overall error balance model.
- Captures non-linearities/interactions from engineered features; dominates in blends/stacking.
- Consistently assigned the largest weight/contribution in advanced modeling approaches.

Feature impact & fairness analysis

- **Age_c** emerged as the top global predictor due to its continuous nature and universal influence across all samples.
- **Smoker status**, while a strong local driver of high charges, impacts only ~20% of the dataset, reducing its overall importance in global models.
- The model prioritizes features that offer consistent, widespread splits (like **age**), rather than rare but extreme effects (like **smoker**).

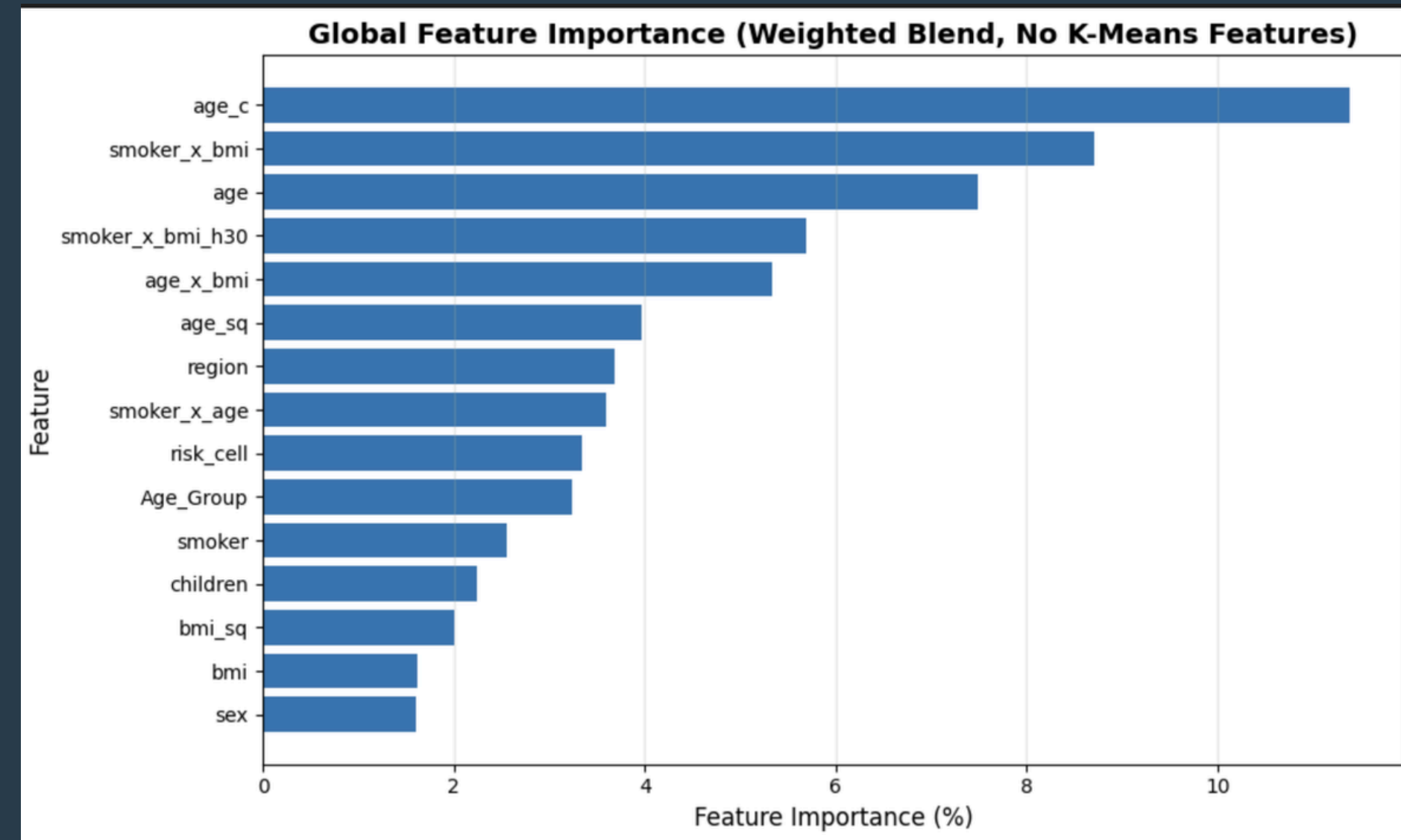


Fig 12: Bar chart of feature importance

Handling Extreme Cases - Model Interpretation with LIME

- Both Instance 44 & 197 have actual charges around 23k, while predictions are 4k, leading severe underestimation.
- Prediction error: Approximately 19k difference in both cases, the two highest error in the dataset.

LIME Interpretation (Feature-Level)

The dominant negative contributors are:

- **smoker_x_bmi_h30** and **smoker_x_bmi**: both $\leq 0 \rightarrow$ model interprets them as non-smokers or low BMI interaction, sharply lowering predicted cost.
- **smoker = 0** $\rightarrow$ further confirms the model assumed non-smoker, heavily reducing charges.
- Contributions from **smoker_x_age** and **age_c**, which slightly adjust cost but cannot offset the smoker penalty like the above.

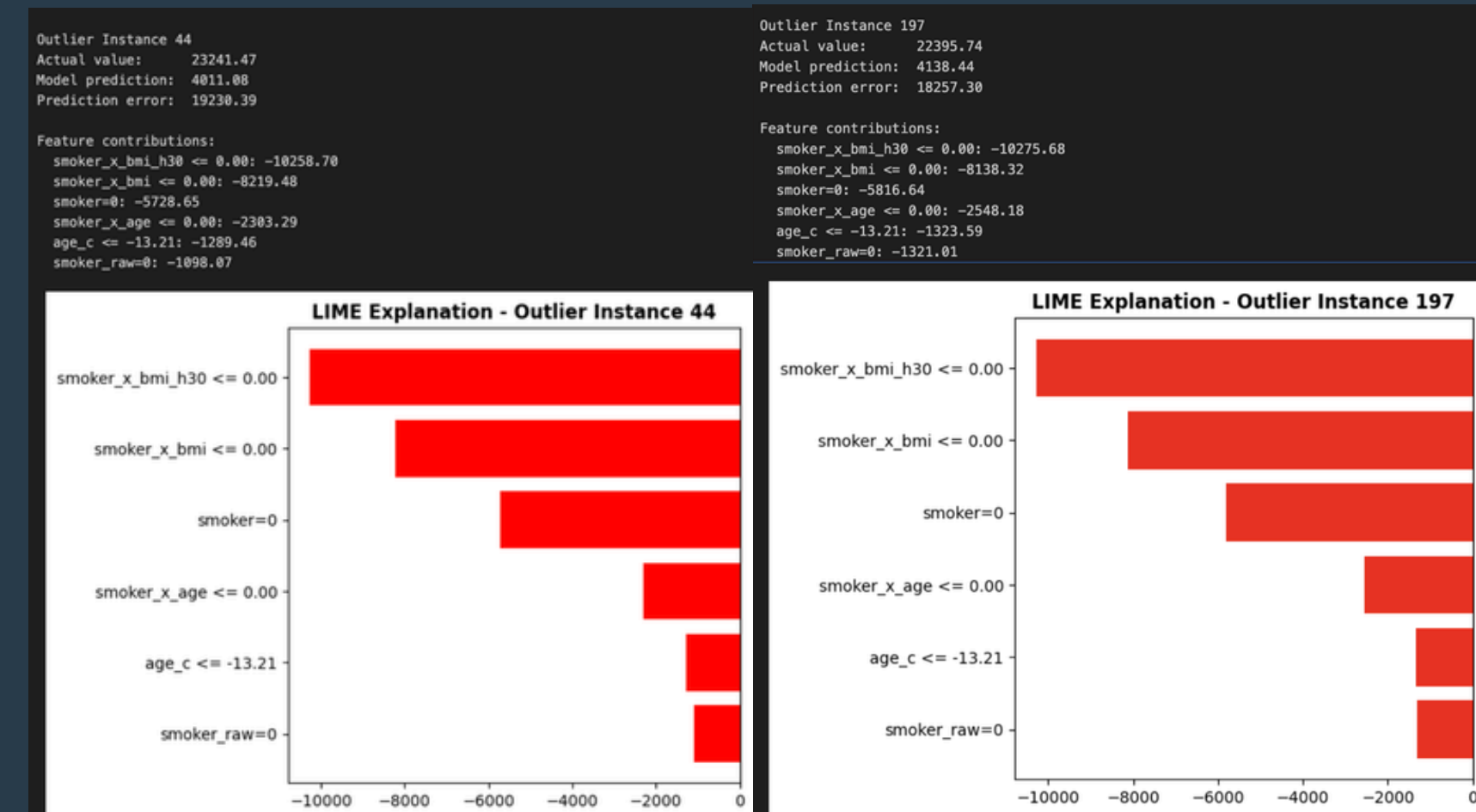


Fig 13: Bar chart of outliers and their features' error contribution

Difficulties faced (Data & Target)

REPRESENTATIVENESS & SIZE

- $n = 1,338$; around 50% BMI ≥ 30 (higher than around 40% US), around 20% smokers (minority class). Risk of sampling bias and imbalance.

TARGET SHAPE

- Charges are right-skewed with extreme outliers (especially smokers) and heteroscedastic variance.

VALIDATION RISK

- Single 80/20 split can be high-variance; tree ensembles (GB/HistGB) can overfit if tuning isn't guarded.

Practical Recommendations

DATASET

- Reweight or stratify: Apply population reweighting (by region/age/smoker) or stratified CV to stabilize segments.
- Balance minority class: Use class/instance weights for smokers (don't upsample for regression unless carefully done).
- Enrich features (future): Add comorbidities, prior claims, income/region SES; confirm BMI measurement timing/accuracy.

MODELING & VALIDATION

- Target handling: Train on $\log(\text{charges})$; report errors on both scales.
- Evaluation: Use nested, stratified K-fold CV; hold out a final lockbox test set.
- Explainability & error analysis: Use SHAP (global + interactions) and residual plots by segment to spot systematic misses.
- Uncertainty: Add conformal prediction or quantile GBM (e.g., 50th/90th) for premium scenarios.

Appendix

JUPYTER NOTEBOOK LINK:

https://drive.google.com/drive/folders/1avRzZbeYoWIXxXI0BhaoCqUIZRS6gbyY?usp=drive_link

References:

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